

5. (2.25 points)

Given the following information about accident year 2017 as of December 31, 2017:

- Accident year 2017 paid loss = \$850,000.
- 2017 earned premium = \$4,000,000.
- Initial expected loss ratio = 67.5%.
- 12-24 month incremental paid link ratio = 1.60.
- 12-ultimate cumulative paid LDF = 3.00.

a. (1.5 points)

Determine the accident year 2017 incremental paid loss in 2018 that would result in the Benktander ultimate loss estimate being \$100,000 less than the Bornhuetter-Ferguson ultimate loss estimate for accident year 2017 as of December 31, 2018. Assume all development factors are unchanged.

b. (0.25 point)

Briefly describe when the Benktander ultimate loss estimate would be greater than the Bornhuetter-Ferguson ultimate loss estimate as of December 31, 2018.

c. (0.5 point)

Explain why it may not be appropriate to use the Bornhuetter-Ferguson method when losses develop downward.

CONTINUED ON NEXT PAGE

EXAM 7 SPRING 2018 SAMPLE ANSWERS AND EXAMINER'S REPORT

SPRING 2018 EXAM 7, QUESTION 5	
TOTAL POINT VALUE: 2.25	LEARNING OBJECTIVE(S): A1, A2
SAMPLE ANSWERS	
Part a: 1.5 points	
<u>Sample 1 – Incremental Paid Formulas</u> BF Ultimate – Benktander Ultimate = 100,000 BF Ultimate = $850 + x + 4,000 \times (0.675) \times [1 - (3.00/1.60)^{-1}] = 2,110 + x$ GB Ultimate = $850 + x + (2,110 + x) \times [1 - (3.00/1.60)^{-1}] = 1,834.67 + 1.467x$ $2,110 + x - 1,834.67 - 1.467x = 100$ $x = 375.45$ <u>Sample 2 – Cumulative Paid Formulas</u> $U_{GB} + 100,000 = U_{BF}$ $24\text{-Ult LDF} = 3.00/1.60 = 1.875$ $BF\ Ult = x + 4,000,000 \times (0.675) \times (1 - 1.875^{-1}) = 1.26M + x$ $Benktander\ Ult = x + (1.26M + x) \times (0.46667) = 1.46667x + 0.588M$ $1.26M - 0.1M + x = 1.46667x + 0.588M$ $x = 1,225,706$ $1,225,706 - 850,000 = 375,706$	
Part b: 0.25 point	
<u>Sample Response</u> • Benktander ultimates would be greater than BF ultimates if the CL Ultimate > BF Ultimate since Benktander is a weighting of the two methods. • Actual paid > expected to date, then the Benktander ultimates are higher • When paid losses emerge higher than expected (in this case, higher than 590k). ○ $(2.11 + x) \times (1 - 1/1.875) + 0.85 + x > 2.11 + x$ ○ $x > 0.590$ • When $U_{BF} > U_0$	
Part c: 0.5 point	
<u>Sample Response</u> • When losses develop downward, the BF method will keep the forward looking IBNR the same, regardless of how losses to date have performed. Thus, the downward development will not affect IBNR. However, in reality, the downward development may be indicating salvage & subro trends that we would also want to apply to our IBNR. • This method selects a priori loss estimate through judgment. If there have been a significant number of claims than usual, this method completely ignores this, even if reported claims > ultimate claims estimated by this method. In these instances, it seems unreasonable to completely ignore such reported claims. • Losses developing downward is unusual and typically the result of favorable closures on claims. Once a claim is closed, it is generally unlikely to reopen and increase further. BF does not recognize this (assumes losses to date not predictive) when, in fact, downward development is generally a good prediction that ultimate losses will be less than	

EXAM 7 SPRING 2018 SAMPLE ANSWERS AND EXAMINER'S REPORT

expected.
EXAMINER'S REPORT
Candidates were expected to calculate unpaid claim estimates using the following loss reserving methods: Budgeted Loss, Chain Ladder, Bornhuetter Ferguson, and Benktander. Candidates were expected to identify and evaluate the strengths and weaknesses and relationships of unpaid claims models at a basic conceptual level by comparing and contrasting these methods, and applying the knowledge of their strengths and weaknesses of these methods in a provided scenario.
Part a
Candidates were expected to calculate unpaid claim estimates using the following loss reserving methods: Budgeted Loss, Chain Ladder, Bornhuetter Ferguson, and Benktander. Common mistakes included: • Selecting an incorrect Link Ratio (deriving the 24-ultimate incorrectly or using the 12-to-ultimate). • Setting up the relationship between the UBF and UGB as given in the problem (e.g. using the wrong sign on the 100k portion of the equation, or dropping the 100k from the relationship altogether). • Reversing the $p = \% \text{Reported}$ vs. $q = \% \text{Unreported}$ in the calculation of ultimates.
Part b
Candidates were expected to identify and evaluate the strengths and weaknesses and relationships of unpaid claims models at a basic conceptual level. Common mistakes included: • Requiring a minimum amount of weight on the Chain Ladder method (e.g. $p = \% \text{reported}$ would need to be greater than 50%). • Stating that losses “come in higher” without a reference comparing to what (e.g. “come in higher than expected”).
Part c
Candidates were expected to know the strengths and weaknesses of the B-F method and justify why it may not be appropriate for the situation in the provided scenario. Common mistakes included: • Not being clear that the BF Reserve or IBNR is independent of the claim emergence. Many candidates stated or implied that the ultimate was independent when in fact claim emergence is a portion of the ultimate calculation. • Citing as facts that negative IBNR is not possible or that $\% \text{Reported}$ cannot exceed 100%, when in fact this is a possible occurrence (e.g. subrogation or salvage at the end of a property damage claim). • Justifying the better method with a generalized phrase without providing an underlying rationale or example (e.g. “we should reflect actual experience more” or “we should rely

EXAM 7 SPRING 2018 SAMPLE ANSWERS AND EXAMINER'S REPORT

upon actual experience less").